

Building CI-CD Pipeline 1 (Dockerisation)

Create Local Docker Files (Dockerised Application)

Docker Files

1. Download the 2215-ssd-10-main from GitHub Repository into Desktop
2. Create 2 Dockerfile at the respective directory from Repo Root

backend/Dockerfile

```
backend/  
├─ Dockerfile  
├─ package.json  
└─ package-lock.json
```

```
FROM node:20-alpine  
  
WORKDIR /app  
  
COPY package*.json ./  
RUN npm ci --omit=dev  
  
COPY . .  
  
EXPOSE 5000  
  
CMD ["npm", "start"]
```

frontend/Dockerfile

```
frontend/  
├─ Dockerfile  
├─ package.json  
└─ package-lock.json
```

```
FROM node:20-alpine AS build  
  
WORKDIR /app  
  
COPY package*.json ./  
RUN npm ci  
  
COPY . .  
RUN npm run build  
  
FROM nginx:alpine  
  
COPY --from=build /app/dist /usr/share/nginx/html  
  
EXPOSE 80  
  
CMD ["nginx", "-g", "daemon off;"]
```

3. Create docker-compose.local.yml in the repo root

```
Windows Laptop  
├─  
├─ VS Code  
├─ Git  
├─ Docker Desktop  
└─ 2216-ssd-10  
    ├─ backend  
    ├─ frontend  
    ├─ Dockerfile  
    └─ docker-compose.local.yaml
```

```
services:  
  mysql:  
    image: mysql:8.0  
    container_name: sitwallet-mysql  
    environment:  
      MYSQL_ROOT_PASSWORD: rootpassword
```

```

    MYSQL_DATABASE: sitwallet

    ports:
      - "3306:3306"

    volumes:
      - mysql_data:/var/lib/mysql

  backend:
    build: ./backend
    container_name: sitwallet-backend
    environment:
      NODE_ENV: production
      HOST: 0.0.0.0
      PORT: 5000
      CORS_ORIGIN: http://localhost:8080
      DB_HOST: mysql
      DB_PORT: 3306
      DB_USER: root
      DB_PASSWORD: rootpassword
      DB_NAME: sitwallet
      JWT_SECRET: replace-this-with-long-random-secret-32chars
      APP_ENCRYPTION_KEY: replace-this-with-valid-key-later
      COOKIE_SECURE: "false"
      CSRF_ENABLED: "true"
    ports:
      - "5000:5000"
    depends_on:
      - mysql

  frontend:
    build: ./frontend
    container_name: sitwallet-frontend
    ports:
      - "8080:80"
    depends_on:
      - backend

volumes:
  mysql_data:

```

4. When the `Dockerfiles` and `docker-compose.local.yml` files created, they should be created at the respective directories

```

Windows Laptop
├──
├── VS Code
├── Git
├── Docker Desktop
└── 2216-ssd-10
    ├── backend
    ├── frontend
    ├── Dockerfile
    └── docker-compose.local.yml

```

5. At the repo root where `docker-compose.local.yml` is located, execute `docker compose -f docker-compose.local.yml up --build` command

The command would do the following:

1. Read `docker-compose.local.yml`
2. Build backend Docker image
3. Build frontend Docker image
4. Download MySQL image
5. Create containers
6. Connect them together
7. Start the application

APP_ENCRYPTION_KEY

6. Before running the command, execute this command first for `APP_ENCRYPTION_KEY`:

```
node -e "console.log(require('crypto').randomBytes(32).toString('hex'))"
```

Command is for generating a valid key in the Powershell and to be used in the application.

`APP_ENCRYPTION_KEY` must be a 32-byte base64 or 64-character hex value.

Copy the value into `docker-compose.local.yml`

```
PS C:\Users\kokse\OneDrive\Desktop\2216-ssd-10-main> docker compose -f docker-compose.local.yaml down
[+] down 4/4
✓Container sitwallet-frontend      Removed    0.0s
✓Container sitwallet-backend      Removed    0.1s
✓Container sitwallet-mysql        Removed    0.0s
✓Network 2216-ssd-10-main.default Removed    0.2s
PS C:\Users\kokse\OneDrive\Desktop\2216-ssd-10-main>
PS C:\Users\kokse\OneDrive\Desktop\2216-ssd-10-main>
PS C:\Users\kokse\OneDrive\Desktop\2216-ssd-10-main> node -e "console.log(require('crypto').randomBytes(32).toString('hex'))"
4197831a39e91d61d532682f41494d1bc1d5a2e34b333ec4e632a97fa5021d6e
971be696c9a1e6cfe7914f107c846fe4d9664459113a7d28834fdfe93ce9c521
PS C:\Users\kokse\OneDrive\Desktop\2216-ssd-10-main> |
```

JWT_TOKEN

- Execute this command first for JWT token:

```
node -e "console.log(require('crypto').randomBytes(32).toString('hex'))"
```

Command is for generating a valid key in the Powershell and to be used in the application.

JWT_TOKEN must be a 32-byte base64 or 64-character hex value.

Copy the value into `docker-compose.local.yaml`

```
PS C:\Users\kokse\OneDrive\Desktop\2216-ssd-10-main>
PS C:\Users\kokse\OneDrive\Desktop\2216-ssd-10-main> node -e "console.log(require('crypto').randomBytes(32).toString('hex'))"
4197831a39e91d61d532682f41494d1bc1d5a2e34b333ec4e632a97fa5021d6e
PS C:\Users\kokse\OneDrive\Desktop\2216-ssd-10-main>
```

```
>Run Service
backend:
  build: ./backend
  container_name: sitwallet-backend
  environment:
    NODE_ENV: production
    HOST: 0.0.0.0
    PORT: 5000
    CORS_ORIGIN: http://localhost:8080
    DB_HOST: mysql
    DB_PORT: 3306
    DB_USER: root
    DB_PASSWORD: rootpassword
    DB_NAME: sitwallet
    JWT_SECRET: 4197831a39e91d61d532682f41494d1bc1d5a2e34b333ec4e632a97fa5021d6e
    APP_ENCRYPTION_KEY: 971be696c9a1e6cfe7914f107c846fe4d9664459113a7d28834fdfe93ce9c521
    COOKIE_SECURE: "false"
    CSRF_ENABLED: "true"
  ports:
    - "5000:5000"
  depends_on:
    - mysql

>Run Service
frontend:
  build: ./frontend
```

`docker-compose.local.yaml`

Creating and Adjusting nginx.conf

- Create `nginx.conf` at the Frontend Folder

`frontend/nginx.conf`

```
server {
    listen 80;
    server_name localhost;

    root /usr/share/nginx/html;
    index index.html;

    location /api/ {
        proxy_pass http://backend:5000/api/;
        proxy_http_version 1.1;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }

    location / {
        try_files $uri /index.html;
    }
}
```

- Add this line into the Dockerfile for frontend

```
COPY nginx.conf /etc/nginx/conf.d/default.conf
```

```
FROM node:20-alpine AS build
```

```
WORKDIR /app
```

```
COPY package*.json ./
```

```
RUN npm ci
```

```
COPY . .
```

```
RUN npm run build
```

```
FROM nginx:alpine
```

```
COPY --from=build /app/dist /usr/share/nginx/html
```

```
COPY nginx.conf /etc/nginx/conf.d/default.conf
```

```
EXPOSE 80
```

```
CMD ["nginx", "-g", "daemon off;"]
```

10. After the changes, execute `docker compose -f docker-compose.local.yaml up --build` command again

11. Make sure that you delete the existing images that are already in the Docker Desktop before executing the `Docker-compose.local.yaml up` command.

Database Setup

12. At this point, the database is not yet initialised with NO TABLES and NO DATA being set up for the application

```
001_schema_mysql57_all_procedures.sql
```

13. Execute the following commands to insert the SQL schema into the Docker Instance:

Command to Inserting Schema:

```
Get-Content .\001_schema_mysql57_all_procedures.sql -Raw | docker exec -i sitwallet-mysql mysql -uroot -prootpassword`
```

Command to Check Schema Within MySQL Database (Docker):

```
docker exec -it sitwallet-mysql mysql -uroot -prootpassword sitwallet -e "SHOW TABLES;"
```

Please take note that the schema was written for MySQL 5.7 and have to align the container to relevant schema.

```
mysql: [Warning] Using a password on the command line interface can be insecure.
PS C:\Users\kokse\OneDrive\Desktop>
PS C:\Users\kokse\OneDrive\Desktop>
PS C:\Users\kokse\OneDrive\Desktop> docker exec -it sitwallet-mysql mysql -uroot -prootpassword sitwallet -e "SHOW TABLES;"
mysql: [Warning] Using a password on the command line interface can be insecure.
+-----+
| Tables_in_sitwallet |
+-----+
| audit_logs          |
| currencies          |
| exchange_rates      |
| login_challenges    |
| platform_settings   |
| recipients          |
| refresh_tokens      |
| roles               |
| transactions        |
| users               |
| wallets             |
+-----+

What's next:
  Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug sitwallet-mysql
  Learn more at https://docs.docker.com/go/debug-cli/
PS C:\Users\kokse\OneDrive\Desktop> docker exec -it sitwallet-mysql mysql -uroot -prootpassword sitwallet -e "SHOW TABLES;|
```

Within the `docker-compose.local.yaml`, the MySQL image have to be adjusted to `5.7` instead of maintaining at `8.0`

```
services:

  mysql:
    image: mysql:5.7
    container_name: sitwallet-mysql
```

Those commands would also have to be executed to adjust the unicode (`utf8mb4_unicode_ci`) within the database instance:

```
Convert DB Collation
docker exec -it sitwallet-mysql mysql -uroot -prootpassword sitwallet -e "ALTER DATABASE sitwallet CHARACTER SET utf8mb4 COLLATE
utf8mb4_unicode_ci;"

Adjust Table (Users) Collation
docker exec -it sitwallet-mysql mysql -uroot -prootpassword sitwallet -e "ALTER TABLE users CONVERT TO CHARACTER SET utf8mb4 COLLATE
utf8mb4_unicode_ci;"
```

```
Adjust Table (audit_logs) Collation
docker exec -it sitwallet-mysql mysql -uroot -prootpassword sitwallet -e "ALTER TABLE audit_logs CONVERT TO CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;"
```

What's next:

```
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug sitwallet-mysql
Learn more at https://docs.docker.com/go/debug-cli/
PS C:\Users\kokse\OneDrive\Desktop> docker exec -it sitwallet-mysql mysql -uroot -prootpassword sitwallet -e "ALTER DATABASE sitwallet CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;"
mysql: [Warning] Using a password on the command line interface can be insecure.
```

What's next:

```
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug sitwallet-mysql
Learn more at https://docs.docker.com/go/debug-cli/
PS C:\Users\kokse\OneDrive\Desktop> docker exec -it sitwallet-mysql mysql -uroot -prootpassword sitwallet -e "ALTER TABLE users CONVERT TO CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;"
mysql: [Warning] Using a password on the command line interface can be insecure.
```

What's next:

```
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug sitwallet-mysql
Learn more at https://docs.docker.com/go/debug-cli/
PS C:\Users\kokse\OneDrive\Desktop> docker exec -it sitwallet-mysql mysql -uroot -prootpassword sitwallet -e "ALTER TABLE audit_logs CONVERT TO CHARACTER SET utf8mb4 COLLATE utf8mb4_unicode_ci;"
mysql: [Warning] Using a password on the command line interface can be insecure.
```

Execute the following commands for restarting docker: `docker restart sitwallet-backend`

To check their unicode, execute the commands:

Check database collation

```
docker exec -it sitwallet-mysql mysql -uroot -prootpassword -e "SELECT SCHEMA_NAME, DEFAULT_CHARACTER_SET_NAME, DEFAULT_COLLATION_NAME FROM information_schema.SCHEMATA WHERE SCHEMA_NAME='sitwallet';"
```

Check table collation

```
docker exec -it sitwallet-mysql mysql -uroot -prootpassword sitwallet -e "SELECT TABLE_NAME, TABLE_COLLATION FROM information_schema.TABLES WHERE TABLE_SCHEMA='sitwallet';"
```

Check Columns collation

```
docker exec -it sitwallet-mysql mysql -uroot -prootpassword sitwallet -e "SELECT TABLE_NAME,COLUMN_NAME,CHARACTER_SET_NAME,COLLATION_NAME FROM information_schema.COLUMNS WHERE TABLE_SCHEMA='sitwallet' AND COLLATION_NAME IS NOT NULL;"
```

```
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug sitwallet-mysql
Learn more at https://docs.docker.com/go/debug-cli/
PS C:\Users\kokse\OneDrive\Desktop> docker exec -it sitwallet-mysql mysql -uroot -prootpassword -e "SELECT SCHEMA_NAME, DEFAULT_CHARACTER_SET_NAME, DEFAULT_COLLATION_NAME FROM information_schema.SCHEMATA WHERE SCHEMA_NAME='sitwallet';"
mysql: [Warning] Using a password on the command line interface can be insecure.
```

SCHEMA_NAME	DEFAULT_CHARACTER_SET_NAME	DEFAULT_COLLATION_NAME
sitwallet	utf8mb4	utf8mb4_unicode_ci

What's next:

```
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug sitwallet-mysql
Learn more at https://docs.docker.com/go/debug-cli/
PS C:\Users\kokse\OneDrive\Desktop> docker exec -it sitwallet-mysql mysql -uroot -prootpassword sitwallet -e "SELECT TABLE_NAME, TABLE_COLLATION FROM information_schema.TABLES WHERE TABLE_SCHEMA='sitwallet';"
mysql: [Warning] Using a password on the command line interface can be insecure.
```

TABLE_NAME	TABLE_COLLATION
audit_logs	utf8mb4_unicode_ci
currencies	utf8mb4_unicode_ci
exchange_rates	utf8mb4_unicode_ci
login_challenges	utf8mb4_unicode_ci
platform_settings	utf8mb4_unicode_ci
recipients	utf8mb4_unicode_ci
refresh_tokens	utf8mb4_unicode_ci
roles	utf8mb4_unicode_ci
transactions	utf8mb4_unicode_ci
users	utf8mb4_unicode_ci
wallets	utf8mb4_unicode_ci

```
PS C:\Users\kokse\OneDrive\Desktop> docker exec -it sitwallet-mysql mysql -uroot -prootpassword sitwallet -e "SELECT TABLE_NAME,COLUMN_NAME,CHARACTER_SET_NAME,COLLATION_NAME FROM information_schema.COLUMNS WHERE TABLE_SCHEMA='sitwallet' AND COLLATION_NAME IS NOT NULL;"
mysql: [Warning] Using a password on the command line interface can be insecure.
```

TABLE_NAME	COLUMN_NAME	CHARACTER_SET_NAME	COLLATION_NAME
audit_logs	action	utf8mb4	utf8mb4_unicode_ci
audit_logs	ip_address	utf8mb4	utf8mb4_unicode_ci
audit_logs	resource_id	utf8mb4	utf8mb4_unicode_ci
audit_logs	resource_type	utf8mb4	utf8mb4_unicode_ci
audit_logs	status	utf8mb4	utf8mb4_unicode_ci
audit_logs	user_agent	utf8mb4	utf8mb4_unicode_ci
audit_logs	user_id	utf8mb4	utf8mb4_unicode_ci
currencies	currency_code	utf8mb4	utf8mb4_unicode_ci
currencies	currency_name	utf8mb4	utf8mb4_unicode_ci
currencies	symbol	utf8mb4	utf8mb4_unicode_ci
exchange_rates	from_currency	utf8mb4	utf8mb4_unicode_ci
exchange_rates	source	utf8mb4	utf8mb4_unicode_ci
exchange_rates	to_currency	utf8mb4	utf8mb4_unicode_ci
exchange_rates	updated_by	utf8mb4	utf8mb4_unicode_ci
login_challenges	challenge_id	utf8mb4	utf8mb4_unicode_ci
login_challenges	created_ip	utf8mb4	utf8mb4_unicode_ci
login_challenges	user_id	utf8mb4	utf8mb4_unicode_ci
platform_settings	description	utf8mb4	utf8mb4_unicode_ci
platform_settings	setting_key	utf8mb4	utf8mb4_unicode_ci
platform_settings	setting_unit	utf8mb4	utf8mb4_unicode_ci
platform_settings	updated_by	utf8mb4	utf8mb4_unicode_ci
recipients	nickname	utf8mb4	utf8mb4_unicode_ci

Please take note if there is an issue with the database:

- 1. Re-import SQL (Get-Content .\001_schema_mysql57_all_procedures.sql -Raw | docker exec -i sitwallet-mysql mysql -uroot -prootpassword)
- 2. Verify tables exist
- 3. Rebuild (docker compose -f docker-compose.local.yaml down / docker compose -f docker-compose.local.yaml up --build)
- 4. Try register again

Step 1: Dockerization	✔ Completed
Step 2: GitHub Actions CI	🕒 Next
Step 3: Build & Push Docker Images	🕒
Step 4: Container Registry (GHCR)	🕒
Step 5: EC2 Auto Deployment	🕒
Step 6: Full CI/CD Demonstration	🕒